

Section 1 already defined the object. These three sections teach you to **take the series apart**, then ask whether what remains is **allowed to be modelled**.

2. Time series decomposition

The working idea is simple: the number you see is not one thing. It is several things stacked.

A useful first split:

$$y_t = \text{Trend} + \text{Season} + \text{Cycle} + \text{Residual}$$

or, when the wiggles grow with the level,

$$y_t = \text{Trend} \times \text{Season} \times \text{Cycle} \times \text{Residual}$$

Decomposition is not a forecasting model. It is a way of **seeing**. After you can see the pieces, you decide whether Holt–Winters, SARIMA, a log, or a difference is the right next tool.

2.1 Trend

Trend is the **slow drift of the level** over a long stretch of time.

- Upward: India's real GDP across decades, a product still gaining customers.
- Downward: a dying SKU, a cost that keeps being engineered down.
- Flat: a mature metric oscillating around a stable mean.

Trend is not the yearly July bump. July is season. Trend is the fact that this July is higher than July five years ago because the whole business grew.

Trend can be roughly linear, curved (fast then flattening), or broken (a policy change, a pandemic, a new plant). Classical decomposition estimates trend with a moving average long enough to wash the season out. STL estimates it with LOESS, which can bend.

If the trend is the only non-stationary piece and it is roughly a straight line, you can subtract that line (de-trend) or difference once. Those two repairs are not identical; we will come

back to that in Section 4.

2.2 Seasonality

Seasonality is a pattern that **repeats with a period you can name from the calendar**.

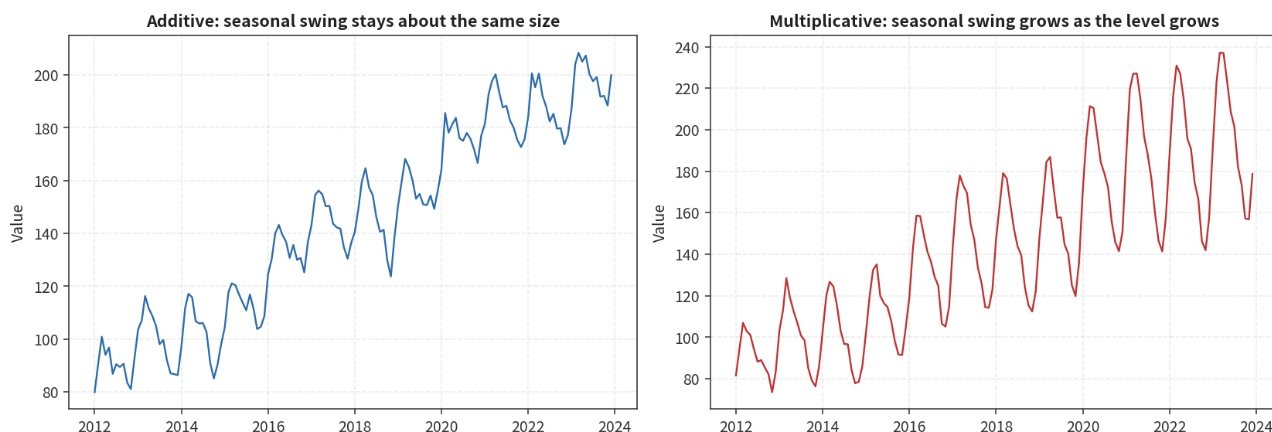
- 24 hours: evening electricity peak.
- 7 days: weekday versus weekend traffic.
- 12 months: ice cream, air-conditioners, airline seats, retail Diwali / Christmas.

The period is known before you model. That is what makes it different from a cycle. You can tell a method: “compare this July to last July.”

Two flavours, and they matter:

- **Additive season:** the July bump is about +20 units every year, whether the series sits at 100 or at 500.
- **Multiplicative season:** July is about 25% above the local trend. As the trend rises, the *rupee* size of the bump grows.

Look at raw plots first. If the waves stay the same height while the series climbs, think additive. If the waves get taller as the level rises, think multiplicative — or take logs and then treat the result as additive.



That single visual is how you choose the decomposition model. Getting it wrong leaks season into the residual or into the trend.

2.3 Cycle — including economic and business cycles

A cycle also rises and falls, but **the period is not fixed to the calendar.**

The textbook cases:

- **Business cycle:** expansion, peak, recession, trough. Sometimes 4 years apart, sometimes 10. You cannot write $s = 8$ and be done.
- **Economic / credit / commodity cycles:** longer swings in investment, credit, or raw-material prices. Same problem: the length moves.

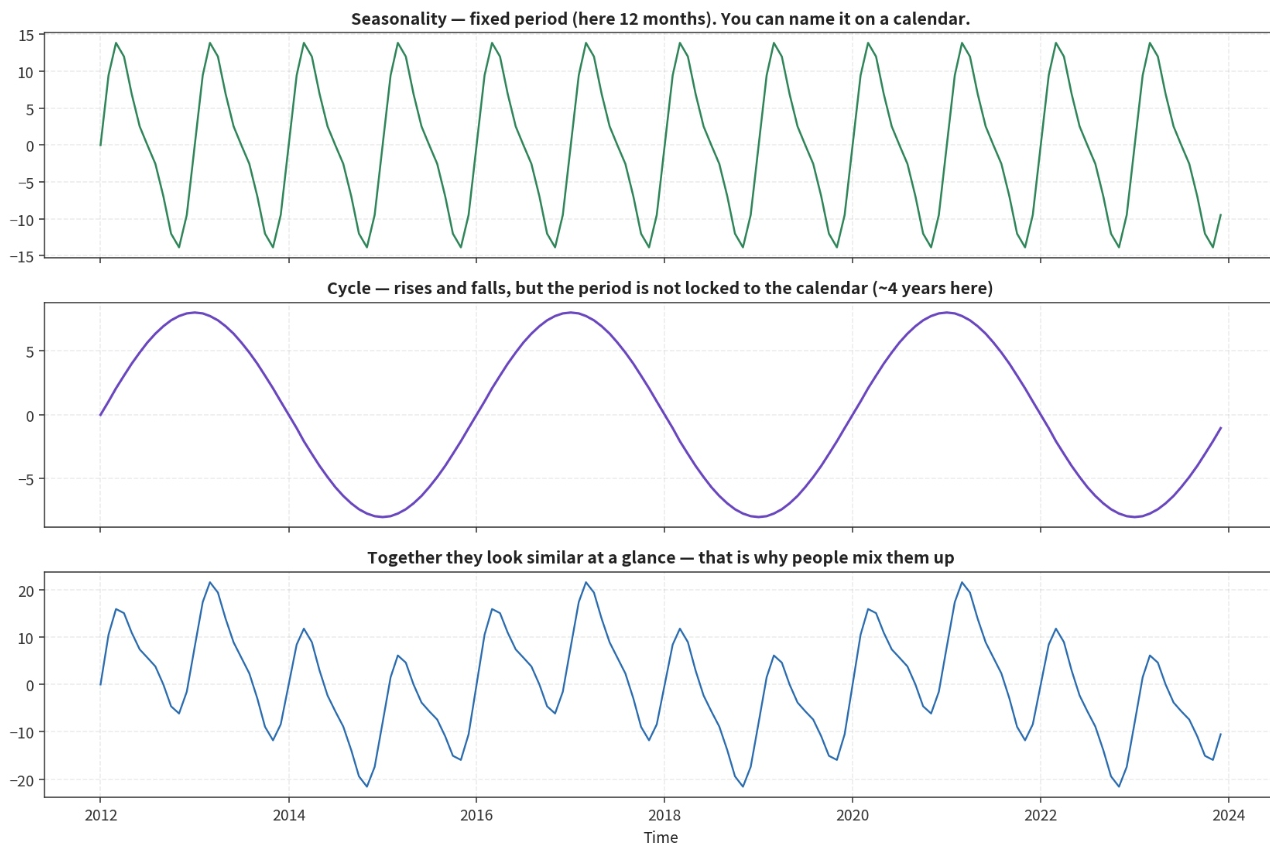
A practical test in your head:

- “This always happens in week 3 of August.” → season.
- “This comes every few years, but I cannot mark the date in advance.” → cycle.

Why the distinction is not pedantry:

- Season can be removed with seasonal dummies, seasonal differences, Holt–Winters seasonal factors, or SARIMA’s $(P, D, Q)_s$ block, because s is known.
- A cycle is usually left inside the residual, or handled with extra explanatory variables (interest rates, an output gap), or with filters that are easy to overuse.

On a short monthly series, a 4-year cycle looks a lot like a slow bend in the trend. That is a real ambiguity. Do not force a “cycle component” just because the slide lists the word.



Top: locked to 12 months. Middle: a slower wave whose peaks do not fall on the same calendar month every time. Bottom: added together they look like “a complicated season.” They are not.

2.4 Residuals

After trend and season (and maybe a cycle) have been taken out, something remains. That leftover is the **irregular / residual**.

The residual is the verdict on the decomposition.

- Ideal: unstructured jitter. No leftover yearly bump, no leftover climb, no obvious change in variance. Then the description is complete enough to hand the leftover to an ARMA, or to treat it as white noise.
- Common failure: a seasonal wiggle still sitting in the residual (wrong period, or additive used on multiplicative data).
- Other failure: a sudden jump (lockdown, strike, promotion) that the trend tried to swallow. That jump is an **intervention**, not noise. Dummy it, or use a robust method (STL with `robust=True`).

Residuals are not “the unimportant part.” They are how you know whether you are finished.

2.5 Additive versus multiplicative — the two classical models

Additive

$$y_t = T_t + S_t + R_t$$

Use when the seasonal swing is roughly constant in *units*. Temperature anomalies, some count series that do not grow, already-logged data.

Multiplicative

$$y_t = T_t \times S_t \times R_t$$

Use when the seasonal swing is roughly constant in *percent*. Sales, passengers, revenue, traffic on a growing site.

A log turns multiplicative into additive:

$$\log y_t = \log T_t + \log S_t + \log R_t$$

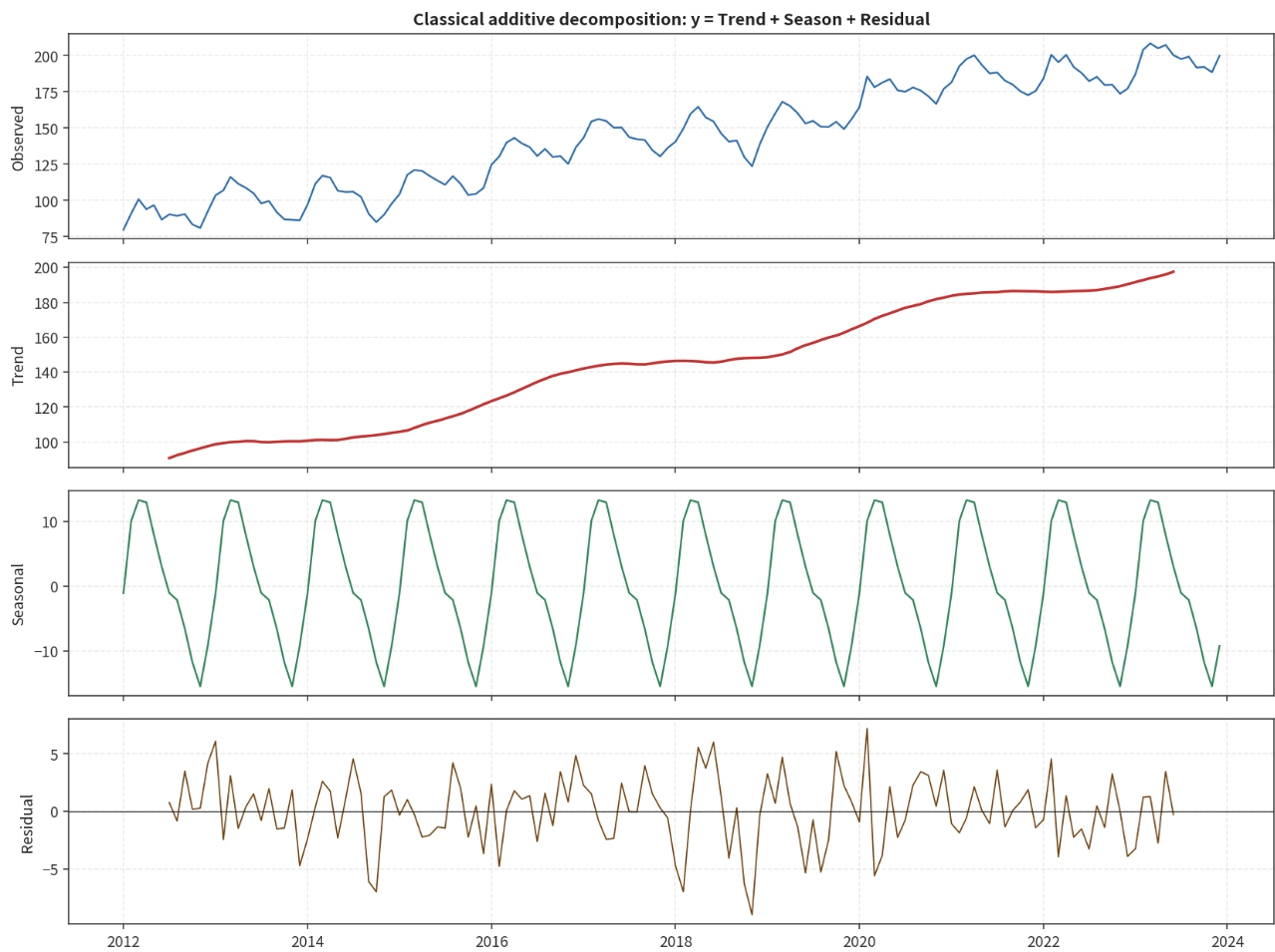
That is why “take logs, then use an additive tool” is a standard move.

Classical decomposition in `statsmodels` estimates T_t with a moving average, averages the de-trended values by month (or by whatever period you named) to get S_t , and calls the rest R_t . It assumes:

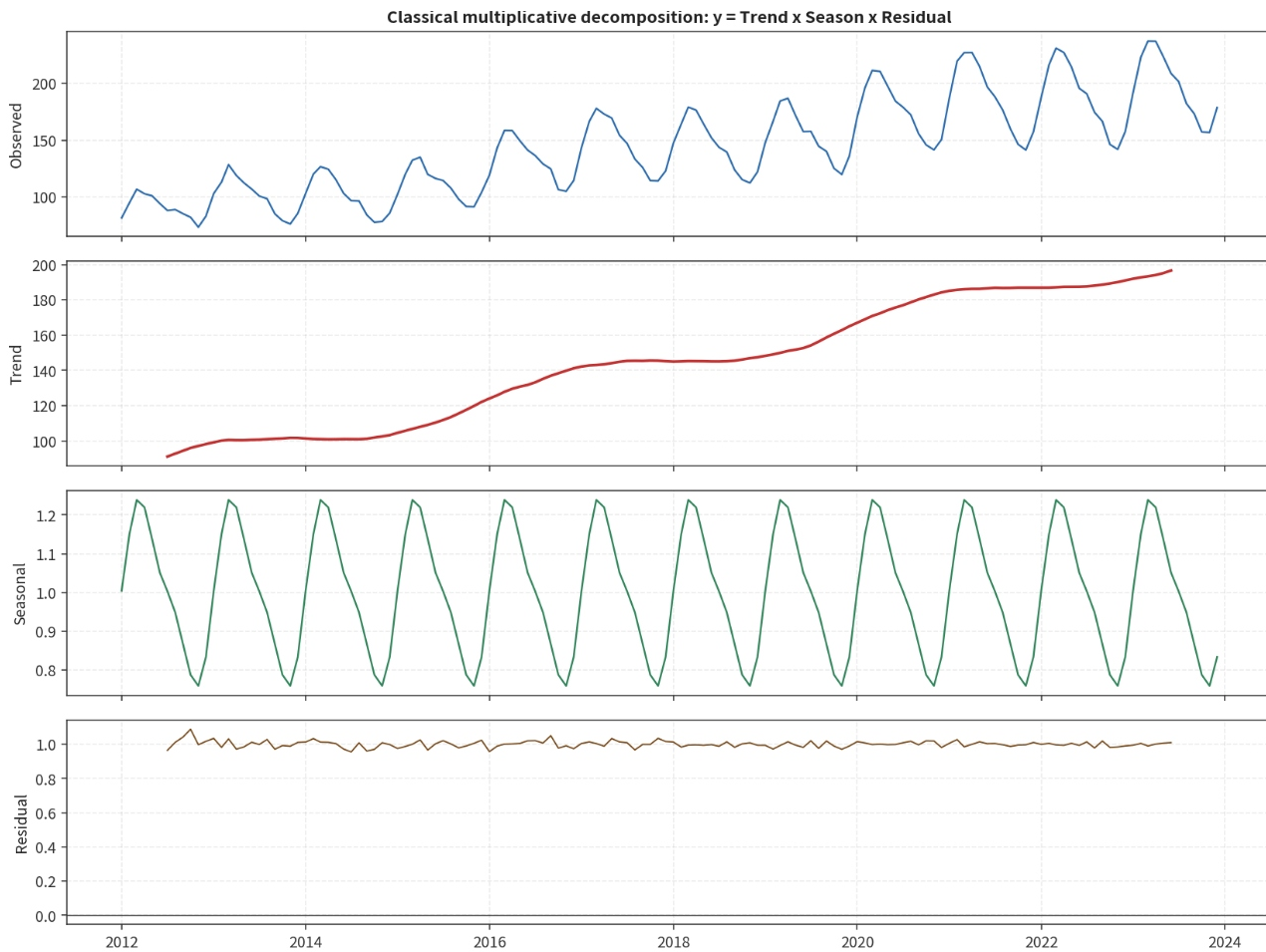
- the seasonal pattern is **stable** across the sample (same January factor every year, in the additive case; same January *ratio* every year, in the multiplicative case),
- the period is known and constant,
- there are enough full seasons to average.

Those assumptions are why STL exists.

Additive decomposition of a series whose season really is additive:



Multiplicative decomposition of a series whose waves grow:



Read any such four-panel figure from the top: raw $\rightarrow$ slow level $\rightarrow$ repeating calendar shape $\rightarrow$ leftover. If the leftover still has a yearly rhythm, the decomposition is wrong.

3. STL — Seasonal and Trend decomposition using LOESS

STL is the modern default when classical decomposition feels stiff.

3.1 What it does

STL uses LOESS (locally weighted scatterplot smoothing) twice:

- a seasonal smoother at lag s (each January is smoothed with other Januarys, each February with other Februarys, ...),
- a trend smoother on the seasonally adjusted series.

It iterates between those two until they stop changing much. The residual is whatever is left

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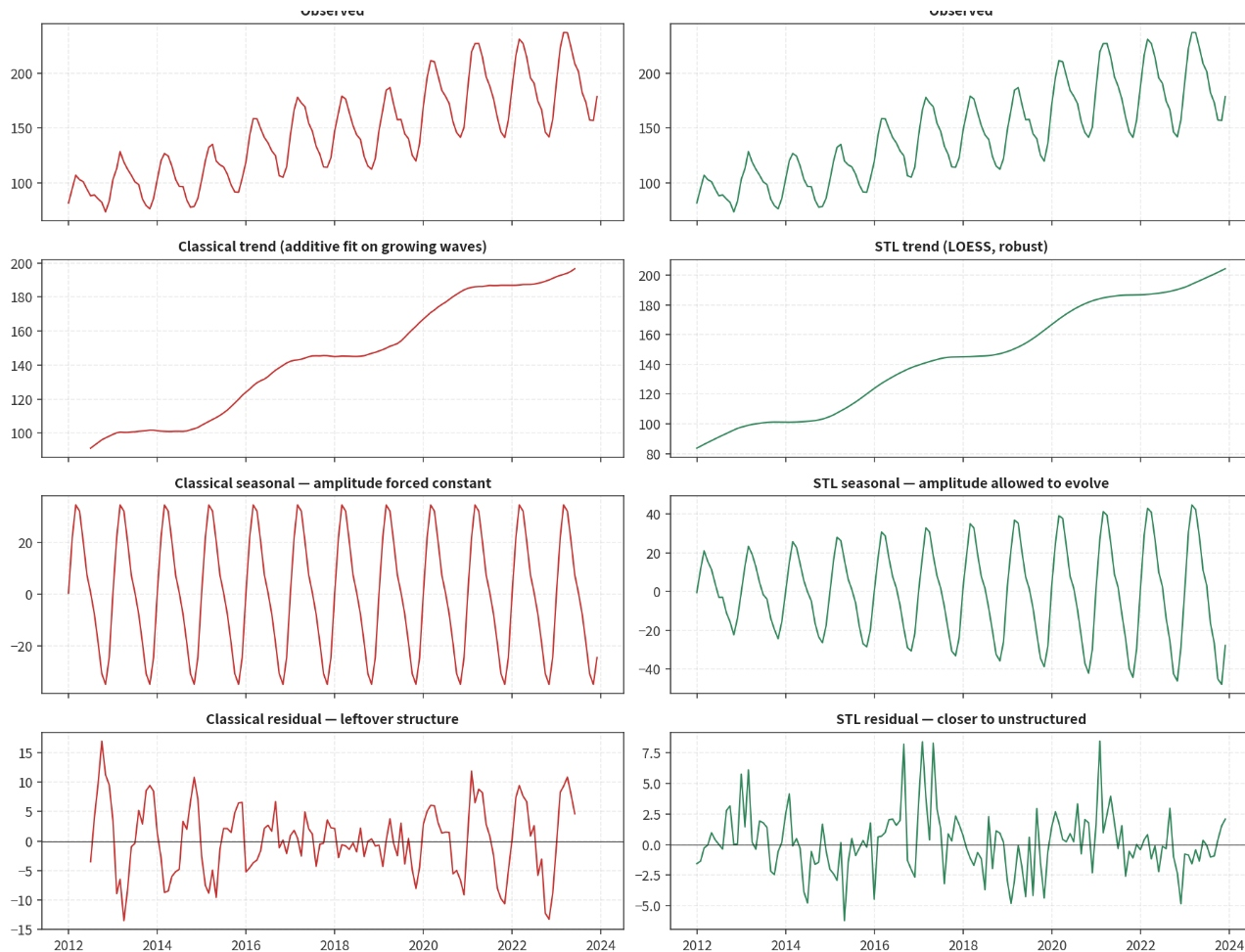
You choose, among other knobs:

- the seasonal period s ,
- how fast the seasonal pattern is allowed to evolve (seasonal smoother window),
- how wiggly the trend is allowed to be (trend window),
- whether to use **robust** weights that down-weight outliers.

3.2 How STL differs from classical decomposition

	Classical	STL
Trend estimator	Moving average of fixed width	LOESS — can bend
Seasonal pattern	Forced almost constant across years	Allowed to evolve slowly
Additive / multiplicative	You pick a model	Native STL is additive. For multiplicative data, decompose $\log y_t$, then exponentiate the pieces
Outliers	Distort trend and season	Robust STL can isolate them in the residual
Multiple seasonalities	Awkward	Still one period per call; for two periods people run MSTL
Calendar quirks (trading days, moving holidays)	Not handled	Not handled either — those need extra regressors
Forecast from the pieces	Not the goal	Not the goal. STL is description. Holt–Winters / SARIMA forecast

The practical difference on a series whose seasonal amplitude grows, if you stubbornly use **classical additive**:



Left column: classical additive on growing waves. The seasonal panel is forced to one height, so leftover structure leaks into the residual.

Right column: STL. The seasonal wave is allowed to get taller. The residual is closer to unstructured. That is the point of the method.

3.3 Choosing the decomposition method

Use this as a default ladder, not a religion.

1. **Plot the raw series.** If waves grow with the level, take logs or use a multiplicative model.
2. If you only need a quick picture and the season looks stable → classical `seasonal_decompose` is fine.
3. If the season is changing, or there are outliers, or the trend is not a gentle curve → **STL (robust)**.
4. If you need a *forecast*, do not stop at STL. Feed the insight into Holt–Winters or SARIMA. STL tells you what is there; it does not by itself produce next year’s July.

5. If there are two calendars at once (hour-of-day *and* day-of-week), look at MSTL or at a model that allows multiple seasonal periods (TBATS, Prophet, a regression with two sets of Fourier terms).

Hyndman & Athanasopoulos treat STL as the standard teaching decomposition in the later editions of *Forecasting: Principles and Practice*. That book is the right companion for this section.

4. Stationarity

This section is why ARIMA has an “I”, why we difference, and why a random walk is not an AR(20).

4.1 Why we need stationarity

A forecasting model estimated on 2015–2020 is being asked to speak about 2021. That only makes sense if the **rules of the series** are the same in both windows.

“Rules” here means the low-order statistics:

- the mean does not depend on the date,
- the variance does not depend on the date,
- the correlation between today and k days ago does not depend on the date — only on k .

If those hold, a model fitted in one stretch is allowed to be used in another. If they fail — the mean is climbing, the variance is exploding, the memory structure is changing — the fitted numbers describe a world that is already gone.

That is the whole reason for Section 4. Not elegance. Transfer of the model across time.

Most of the classical family (AR, MA, ARMA, VAR) **assumes weak stationarity**. ARIMA’s extra step is: change the series until that assumption is plausible, then model the changed series.

4.2 Two kinds of stationarity

Strict (strong) stationarity.

The entire joint distribution of $(y_t, y_{t+1}, \dots, y_{t+k})$ does not depend on t . Every moment, every tail, every dependence structure is time-invariant.

This is almost never testable in applied work. We do not have enough data to see the whole distribution at every date.

Weak (second-order, covariance) stationarity.

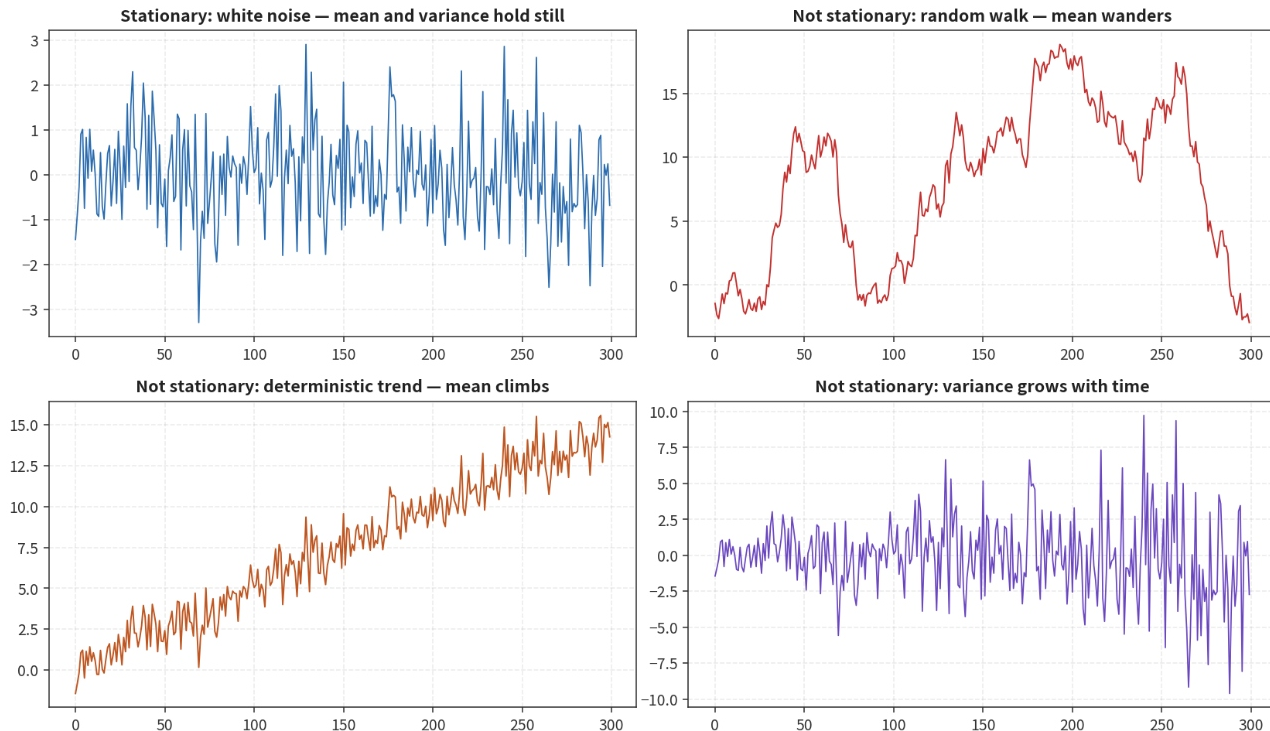
Only the first two moments are required to be stable:

1. $\mathbb{E}(y_t) = \mu$ finite, same for all t .
2. $\text{Var}(y_t) = \sigma^2$ finite, same for all t .
3. $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ depends on the lag k , not on t .

This is the definition used by ARMA theory, by ACF/PACF, and by ADF/KPSS. When a course says “stationary,” it means this unless it says otherwise.

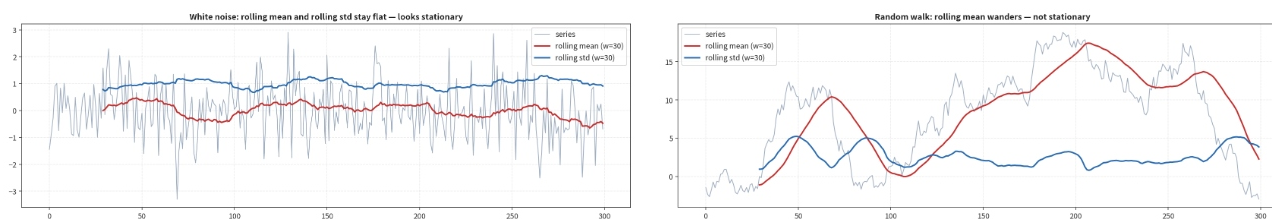
A Gaussian weakly stationary series is also strictly stationary, because a Gaussian law is fixed by mean and covariance. That is why textbooks often slide between the two words.

Four pictures of the idea:



Only the top-left panel is a process you may feed straight to ARMA. The other three break at least one of the three weak-stationarity conditions.

A rolling mean and rolling standard deviation make the same point without a test statistic:



If those rolling lines drift, do not argue with the plot. The series is not weakly stationary.

4.3 Testing weak stationarity: ADF and KPSS

You already use the plot. The tests are a second opinion. They disagree on purpose, because their nulls are opposite.

Augmented Dickey–Fuller (ADF)

Null hypothesis:

H_0 : the series has a unit root (random-walk-like, not stationary)

Alternative: the series is stationary (around a mean, or around a trend, depending on which version you run — “none”, “intercept”, “intercept + trend”).

You reject H_0 when the ADF statistic is **sufficiently negative** and the p -value is small. Rejecting ADF is evidence *for* stationarity.

“Augmented” means extra lags of Δy_t are put on the right-hand side so leftover short-run correlation does not fool the test. Those lags are usually chosen by AIC.

Trap: ADF has low power. A very persistent stationary AR(1) with $\phi = 0.97$ looks a lot like $\phi = 1$. ADF will often fail to reject. “Fail to reject unit root” is not the same as “prove a unit root.”

KPSS

Null hypothesis:

H_0 : the series is (trend-)stationary

Alternative: there is a unit root.

You reject H_0 when KPSS is large and the p -value is small. Rejecting KPSS is evidence *against* stationarity.

Which one to choose?

Use **both**. The interesting cases are the disagreements.

ADF	KPSS	Reading
Reject unit root	Fail to reject stationarity	Comfortably stationary
Fail to reject unit root	Reject stationarity	Treat as unit root; difference
Fail to reject unit root	Fail to reject stationarity	Under-powered sample; look at the plot, consider a first difference anyway or get more data
Reject unit root	Reject stationarity	Something else is wrong: break, changing variance, season, or you used the wrong deterministic terms (constant / trend)

Always match the test specification to what you can see. If the series clearly climbs, run ADF with trend, and KPSS with trend. Testing a trending series for stationarity *around a constant* will call it non-stationary for a boring reason.

There is no serious everyday test for **strict** stationarity. People sometimes look at whether higher moments or the full shape change across halves of the sample. For this course, weak stationarity plus a plot is the job.

4.4 Making a series stationary

You have four families of repair. Use the lightest one that works. Over-differencing turns a stationary series into an MA with a unit root and makes forecasts worse.

Differencing

First order:

$$\Delta y_t = y_t - y_{t-1}$$

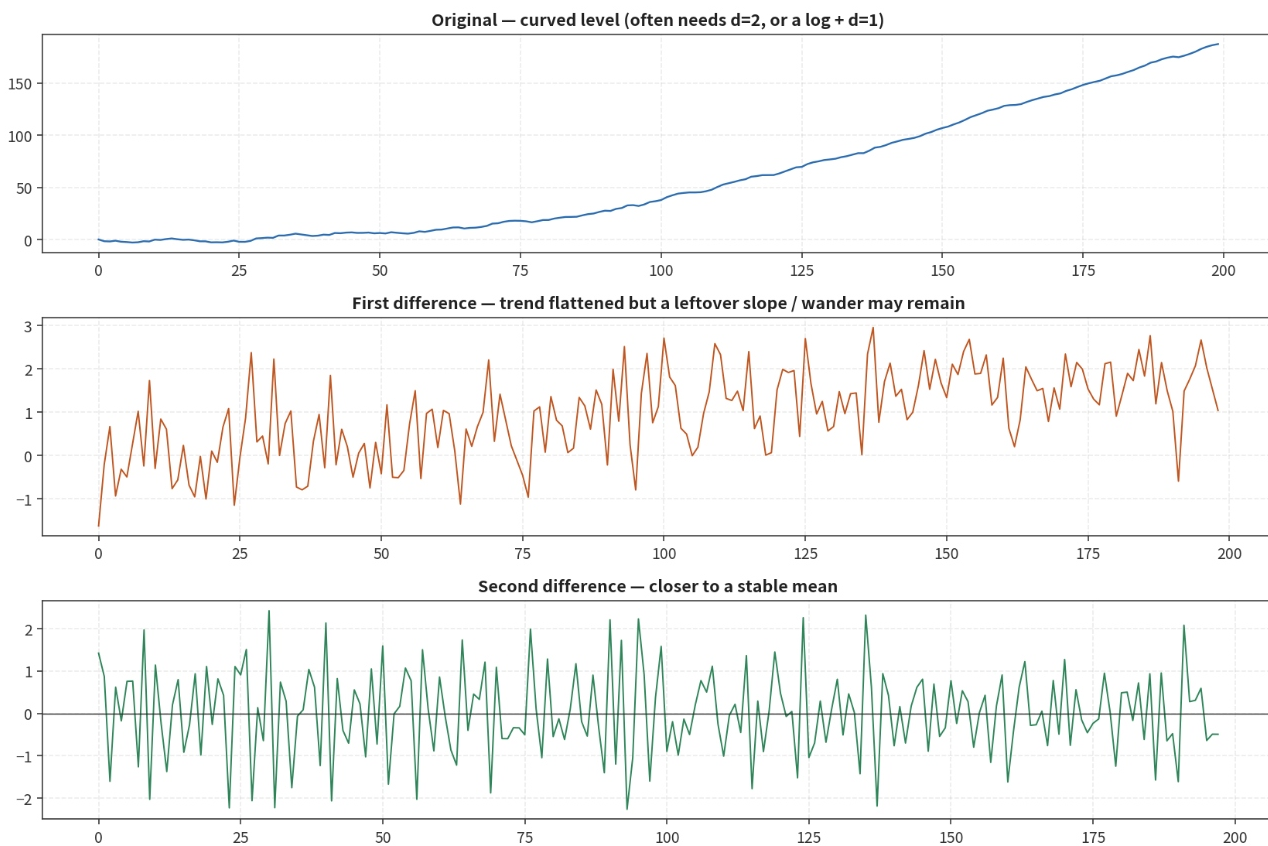
$$\Delta y_t = y_t - y_{t-1}$$

This kills a random walk and a linear stochastic trend. It is the “I” in ARIMA when $d = 1$.

Second order:

$$\Delta^2 y_t = \Delta y_t - \Delta y_{t-1} = y_t - 2y_{t-1} + y_{t-2}$$

This is for a series whose *slope* itself wanders (integrated of order 2). Rare in monthly business data. Common warning sign: you thought you needed $d = 3$. Usually you needed a log, or a break dummy, not a third difference.



Top: curved level. Middle: first difference still has a leftover drift. Bottom: second difference sits on a stable mean. That is what $d = 2$ is for. Do not start there; arrive there only if $d = 1$ is visibly not enough.

Seasonal differencing is the same idea at lag s :

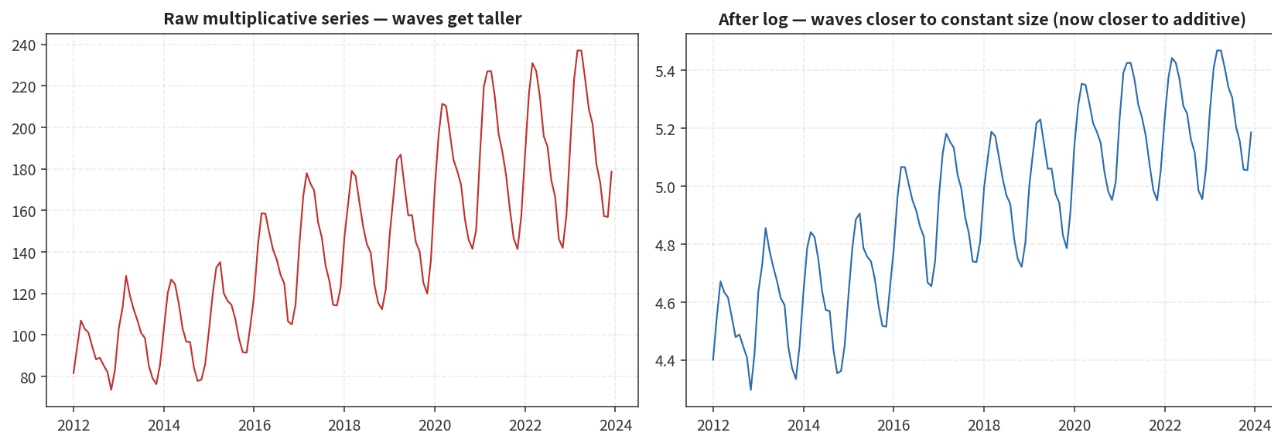
$$\Delta_s y_t = y_t - y_{t-s}$$

Monthly data with a yearly wave often wants $\Delta\Delta_{12}$, i.e. $d = 1$ and $D = 1$.

Transformations

These attack **variance** and multiplicative season more than they attack a wandering mean.

- **Log:** $y_t > 0$. Turns “wiggle grows with level” into a more stable percentage wiggle. Also turns multiplicative season into additive



- **Power:** square root, cube root, etc., when logs are too strong or zeros are present (logs hate zeros).
- **Box-Cox:** a family that includes log and powers,

$$y_t^{(\lambda)} = \begin{cases} (y_t^\lambda - 1)/\lambda & \lambda \neq 0 \\ \log y_t & \lambda = 0 \end{cases}$$

Estimate λ from the data (often to stabilise variance). $\lambda = 1$ is “do nothing.” $\lambda = 0$ is log.

A transform does not remove a unit root by itself. You often **log first, then difference**.

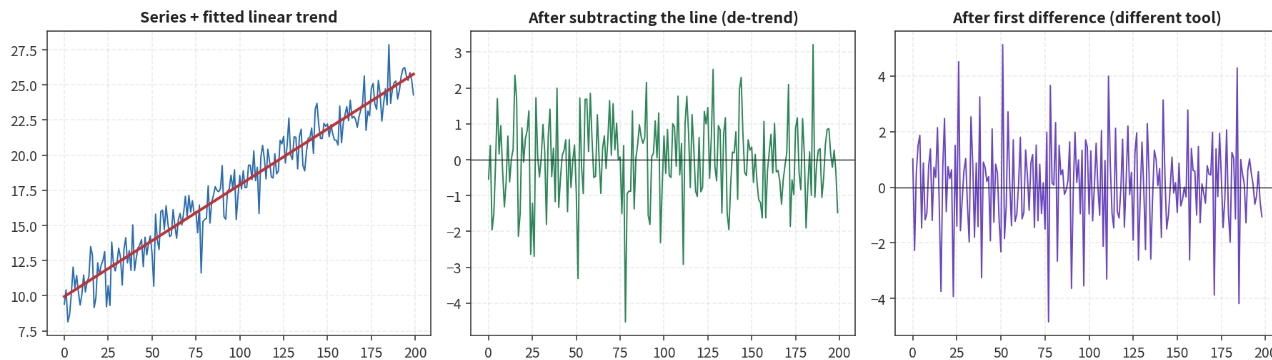
De-trending

If the non-stationarity is a **deterministic** line (or a smooth curve) rather than a random walk, you can subtract that line.

- Linear: regress y_t on t , keep the residual.
- Moving-average de-trend: subtract a long rolling mean.

Differencing a deterministic linear trend also works — $\Delta(a + bt) = b$ plus the differenced noise — but it is a blunter tool. Differencing a stationary series that only had a line under it produces an MA structure you then have to model. If you *know* the trend is a straight line

produces an IMA structure you then have to model. If you know the trend is a straight line, subtracting the line is cleaner.



Left: series and the fitted line. Middle: subtract the line; leftover is jitter around zero. Right: first difference of the same series. Both can look “stationary.” They are not the same model of the world. ADF with a trend term is how you ask which world you are in.

Seasonal adjustment

Sometimes the only non-stationarity you care about is the calendar wave. Seasonal adjustment (X-13, STL then subtract / divide the seasonal factor, dummy variables for months) produces a seasonally adjusted series that is easier to test and to model.

If the goal is forecasting *with* the season, do not throw the season away. Put it in Holt–Winters or SARIMA. Seasonal adjustment is for description, for comparing months, and for models that cannot speak season themselves.

4.5 Choosing the right repair

A working order that matches this curriculum:

1. Plot. Name trend, season, changing variance, breaks.
2. If waves grow with the level → log or Box–Cox.
3. If there is a calendar wave you intend to model → keep it for SARIMA / Holt–Winters, or seasonally difference ($D = 1$).
4. Rolling mean / rolling std. ADF + KPSS, with the right deterministic terms.
5. If tests and plot say unit root → first difference. Look again.
6. If the differenced series still climbs → consider $d = 2$, but also consider that you wanted a log or a broken trend.

7. If the series is clearly a straight line plus stationary noise → de-trend, do not automatically difference.
8. Stop as soon as the repaired series has a stable rolling mean and std, a quickly dying ACF, and ADF/KPSS that agree it is stationary.
9. Then, and only then, read ACF/PACF for p and q .

Over-repair is a real mistake. Two extra differences create artificial MA patterns and throw away long-run level information you may have wanted.

How 2, 3 and 4 connect to the models you already have

Decomposition (2, 3) answers: *what is in this series?*

Stationarity (4) answers: *is the leftover, or the raw series, allowed into ARMA?*

ARIMA's d and SARIMA's D are the stationarity repairs turned into model orders.

Holt–Winters is a forecasting system that **does not require you to difference first**; it carries trend and season as state variables instead. That is why smoothing and ARIMA can both be right on the same series.

A short pipeline you can actually run:

```
text
```

```
plot
```

- decompose (classical or STL) to see T / S / R
- decide additive vs multiplicative (or log)
- test stationarity on the series you will model
- difference / de-trend / seasonally adjust as needed
- ACF / PACF → ARIMA / SARIMA
- or skip the last two and use Holt–Winters if T and S are the whole story

Resources that match these three sections

- Hyndman & Athanasopoulos, *Forecasting: Principles and Practice* — chapters on time series decomposition and on stationarity / differencing. Free online; the best single companion.
 - Cleveland et al., the original STL paper, if you want the smoother details.
 - `statsmodels.tsa.seasonal.seasonal_decompose` and `STL`.
 - `statsmodels.tsa.stattools.adfuller` and `kpss`. Read the docstring for the `regression` argument ('c' vs 'ct').
 - For the unit-root idea in one picture: the three-panel AR(1) with $\phi = 0.5, 0.9, 1.0$ from the previous lesson.
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Next, in the same style, the highest-value follow-ups are a **worked airline-passengers (or your series) pipeline**: STL + ADF/KPSS + $\Delta\Delta_{12}$ + ACF/PACF + a small SARIMA next to Holt–Winters. Save if you want that on a public series or on data you will paste